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Experimental investigation and thermodynamic assessment of the Fe-Pr and Fe-Nd binary systems

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ABSTRACT

In this work, Fe-Pr alloys and Fe-Nd alloys were investigated experimentally by means of thermal analysis. The temperatures of the invariant reactions and liquidus in the Fe-Pr and Fe-Nd binary systems were measured. Based on the experimental results determined in this work and the critical review of the available experimental data in published literature, the Fe-Pr and Fe-Nd binary systems were re-assessed thermodynamically using the CALPHAD method. The solution phases including liquid, bcc, fcc and dhcp are modeled by the substitutional solution model and their excess Gibbs energies are expressed with the Redlich-Kister polynomial. The intermetallic compounds, $Fe_{17}Pr_2$, $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$ are treated as intermetallic compounds. The self-consistent thermodynamic parameters obtained finally to describe the Gibbs energies of various phases in the Fe-Pr and Fe-Nd binary systems can be used to reproduce well the phase equilibria and thermodynamic data.

1. Introduction

Nd-Fe-B permanent magnets with excellent magnetic properties have been used widely in many industrial applications, such as wind power, electric vehicle and hard disk [1–6]. In general, higher energy product, coercivity and temperature stability of $Nd_2Fe_{14}B$-based permanent magnets have been achieved through the addition of heavy rare earth (RE) elements (e.g. Dy, Tb) in Nd-Fe-B magnets [7,8]. To decrease the usage of expensive heavy rare earth elements, light rare earth elements (e.g. Ce, Pr) could be used to substitute the partial content of heavy rare earth elements in Nd-Fe-B permanent magnets. Meanwhile, the addition of light rare earth elements (e.g. Ce, Pr) in Nd-Fe-B permanent magnets has to be controlled to meet the requirement of good magnetic properties [9–12]. In order to further understand the effect of light rare earth elements on magnetic properties of Nd-Fe-B permanent magnets, phase equilibria and thermodynamic information of the RE-Fe-B systems are indispensable.

As the key sub-binary systems, the Fe-Pr and Fe-Nd binary systems were assessed by Bär and Schaller [13] and Hennmann et al. [14], respectively. Marazza et al. [15] reviewed these two binary systems. Based on the calculated mixing enthalpy of liquid Fe-Pr alloys using the Miedema method, Zhou and Zeng [16] optimized the Fe-Pr binary system. Using the modified quasi-chemical model to describe liquid phase, the Fe-Nd binary system was re-optimized thermodynamically by Ende et al. [17]. In the evaluation of the Fe-Pr and Fe-Nd binary systems, the present authors [18] found that the published thermodynamic descriptions of these two binary systems [13,14,16,17] could not reproduce satisfactorily the latest experimental enthalpies of mixing of liquid alloys at 1829 K measured by Ivanov et al. [19]. Recently, considering the latest experimental results [19], thermodynamic calculations of the Fe-Pr and Fe-Nd binary system were updated by Konar et al.[20] using the modified quasi-chemical model to describe liquid phase. Although the calculated results are in good agreement with the experimental results including the latest enthalpies of mixing of liquid alloys [19], the intermatallic compound, $Fe_{17}Pr_2$, would decompose at 632 K and is not stable at room temperature. Thus, the thermal stability of the intermetallic compounds and invariant reactions in the Fe-Pr and Fe-Nd binary systems would be determined in further experimental verification. In addition, considering the compatibility of thermodynamic database of the RE-Fe-B multicomponent systems, experimental investigation and thermodynamic re-assessment of the Fe-Pr and Fe-Nd binary systems are still necessary using the Thermo-calc® software.

In the present work, the Fe-Pr and Fe-Nd binary systems were firstly investigated experimentally by means of differential thermal analysis (DTA), and then were assessed thermodynamically based on critical review of the reported experimental phase equilibria data and thermodynamic data using the CALPHAD method.

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2. Literature information

2.1. The Fe-Pr binary system

The Fe-Pr binary phase diagram was firstly investigated experimentally by Ray [21]. Using thermal analysis and X-ray method, Tian et al. [22] studied the Fe-Pr binary phase diagram in the composition range from $Fe_{17}Pr_2$ to Pr and confirmed $Fe_{17}Pr_2$ as a stable intermetallic compound. Furthermore, Zhuang et al. [23] measured the Fe-Pr binary phase diagram using differential scanning calorimetry and X-ray diffraction. Their experimental results [23] including the phase relations and temperatures of invariant reactions are obvious difference with the previously reported results [13,21,22]. Du et al. [24] used only the experimental data [23] to optimize the Fe-Pr binary phase diagram. In the assessment of the Fe-Pr binary system, Zhang et al. [25] recommended two peritectic reactions, L+δ-Fe↔γ-Fe at 1671 K and L+γ-Fe↔$Fe_{17}Pr_2$ at 1378 K, the peritectoid reaction, γ-Fe +$Fe_{17}Pr_2$↔α-Fe at 1189 K, and the eutectic reaction, L↔$Fe_{17}Pr_2$+α-Pr at 940 K.

In the Fe-Pr binary system, $Fe_{17}Pr_2$ and Fe_2Pr as the stable intermetallic compounds were reported by Zhuang et al. [23]. Burkhanov et al. [26] reported $Fe_{17}Pr_2$ is a stable intermetallic compound, while Fe_2Pr is a metalstable compound. Subsequently, Burkhanov et al. [27] suggested Fe_2Pr is stable above 893 K. Based on the experimental results determined by single-crystal X-ray diffraction method [28], powder X-ray spectra [29] and thermal analysis [22], $Fe_{17}Pr_2$ is the only stable intermetallic compound at the standard condition (298.15 K, 1 atm), while Fe_2Pr is stable under high pressure and thus was not be observed by Tian et al. [22] and Bär and Schaller [13]. In the review of the Fe-Pr binary system, Okamoto [30–32] suggested only $Fe_{17}Pr_2$ is a stable intermetallic compound.

The enthalpies of mixing and the partial enthalpies of Pr and Fe in the Fe-Pr liquid alloys at 1829 K were determined by Ivanov et al. [19] using an isoperibolic calorimetry. The activities of Pr in the Fe-Pr alloys at 923 K, 973 K, 1023 K and 1073 K were determined by Bär and Schaller [13] by means of the electromotive force (EMF) method. Based on EMF results, the enthalpy of formation of $Fe_{17}Pr_2$ was determined to be −1.92 kJ/mole-atom at 973 K [13], which shows obvious difference with the data (−0.2 ± 3.0 kJ/mole-atom) measured by Meschel et al. [33] using high temperature direct synthesis calorimetry. Meschel et al. [33] also calculated the enthalpy of formation of $Fe_{17}Pr_2$ (−3.2 kJ/mole-atom) with the first principle calculation. Using the EMF method, Gozzi et al. [34] determined the enthalpy and entropy of formation of $Fe_{17}Pr_2$ at 995 K to be 1.09 ± 0.02 kJ/mole-atom and 2.24 ± 0.01 JK^{-1}/mole-atom, respectively, and the Gibbs energy of formation of $Fe_{17}Pr_2$ at 995 K was calculated to be −1.141 ± 0.001 kJ/mole-atom.

In addition, Bär and Schaller [13] determined the Curie temperature and magnetic moment of $Fe_{17}Pr_2$ to be 283 K and 30.6 μ_B, respectively. Using PPMS measurements, Mandal et al. [35] measured the heat capacity and Curie temperature (278 K) of $Fe_{17}Pr_2$.

2.2. The Fe-Nd binary system

The Fe-Nd binary system was studied experimentally by Terekhova et al. [36], Che et al. [37], Schneider et al. [38] and Faudot et al. [39] using thermal analysis and X-ray diffraction method. It was found in Refs. [36–39] that the intermetallic compound, $Fe_{17}Nd_2$ is only stable. The Fe-Nd binary system was reviewed by Zhang et al. [40] and shows a stable intermetallic compound, $Fe_{17}Nd_2$. However, by means of thermal analysis and X-ray diffraction method, Landgraf et al. [41] discovered that the intermetallic compound, $Fe_{17}Nd_5$ is also stable and thus revised the Fe-Nd binary phase diagram. Hennmann et al. [14] determined experimentally the Fe-Nd binary phase diagram by means of DTA, mass spectrometry and X-ray diffraction and $Fe_{17}Nd_5$ was confirmed to be stable. Using the single crystal X-ray diffraction method, the lattice parameters of $Fe_{17}Nd_5$ were determined by Moreau et al. [42]. The crystal structure information and magnetic characterization of $Fe_{17}Nd_5$ were investigated by Lin et al. [43] and Landgraf et al. [44] through magnetic measurements and neutron diffraction method. Considering two stable intermetallic compounds, $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$, Hallemans et al. [45] optimized the Fe-Nd binary system in the calculation of the Nd-Fe-B ternary system. The review of the Fe-Nd binary system was updated by Okamoto [46,47].

In addition, Zhang et al. [48] determined the solubility of Nd in α-Fe to be 0.0213 at% Nd at 298 K with the X-ray diffraction method. Using the microprobe quantitative analysis and X-ray diffraction method, Li et al. [49] measured the solubility of Nd in α-Fe in the temperature range of 873–1153 K. The solubility of Nd in Fe-Nd alloys at 1073 K was determined to be 0.47 ± 0.05 at% Nd through the measured lattice parameters of alloy samples in the Fe-rich side. On the other hand, the Curie temperature and the effective magnetic moment of $Fe_{17}Nd_2$ reported by Hennmann et al. [14] are 327 K and 2.3 μ_B, respectively. The Curie temperature of $Fe_{17}Nd_5$ was determined to be 503 K by Landgraf et al. [44].

Using the isoperibolic calorimetry, Ivanov et al. [19] measured the enthalpy of mixing and partial enthalpies of Fe and Nd in the Fe-Nd liquid alloys at 1829 K. With double Knudsen cell mass spectrometry and EMF measurements, Hennmann et al. [14] and Nagai et al. [50] determined the activities of Nd in the Fe-Nd alloys at 973 K, 1073 K and 1373–1523 K, respectively. Hennmann et al. [14] determined the Gibbs energies of formation and heat capacities of these two intermetallic compounds, $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$. Based on the activities of Nd in the Fe-Nd alloys, Nagai et al. [50] obtained the Gibbs energy of formation of $Fe_{17}Nd_2$ at 1373–1523 K. Using EMF measurements, the enthalpy of formation of $Fe_{17}Nd_2$ at 1022 K and the partial excess Gibbs energy of $Fe_{17}Nd_2$ were determined by Gozzi et al. [34]. The enthalpy of formation of $Fe_{17}Nd_2$ at 1373 K was determined by Meschel et al. [33] using the high temperature direct synthesis calorimetry, which shows good agreement with their ab-initio calculation.

3. Experiments and results

Bulk Fe, Pr and Nd (99.99% purity, China New Metal Material Technology Company, Ltd.) were used as the starting materials. Ingots of Fe-Pr and Fe-Nd alloys with the mass of about 2 g were melted three times in an arc furnace by arc melting pure elements using a non-consumable tungsten electrode under an inert argon atmosphere. No chemical analysis for the alloys was conducted because the mass lost is generally less than 1% during arc melting. To achieve composition homogeneity, the Fe-rich alloys (Fe92.5Pr7.5, Fe92.5Nd7.5, Fe85Nd15) and the other Fe-Pr and Fe-Nd alloys were sealed in evacuated quartz tubes under vacuum (< 10^3 Pa) to be annealed at 873 K for 240 h and 773 K for 168 h, respectively, and then quenched into ice water. Alloys samples were examined by differential thermal analysis (DTA, TA Instruments SDT/Q-600) at different heating and cooling rates of 10 and 20 K/min using Al_2O_3 crucibles under a flowing pure Ar atmosphere Thermal analysis measurements were calibrated using calibration materials In, Sn, Bi, Zn, Al, Ag, Au and Ni (TA Instruments-Waters LLC) as standard samples in order to eliminate the random errors and systematic errors. According to DTA measurements, onset temperatures of the peaks in the DTA curves of alloy samples were treated as the temperatures of the invariant reactions and the peak temperature of the last peak was selected to be the liquidus temperature.

3.1. The Fe-Pr binary system

Fig. 1 shows the representative DTA curves of the as-cast and annealed Fe-Pr alloys measured in this work. As shown in Fig. 1(a), two peaks in the DTA curve of the annealed Fe77Pr23 alloy are obviously

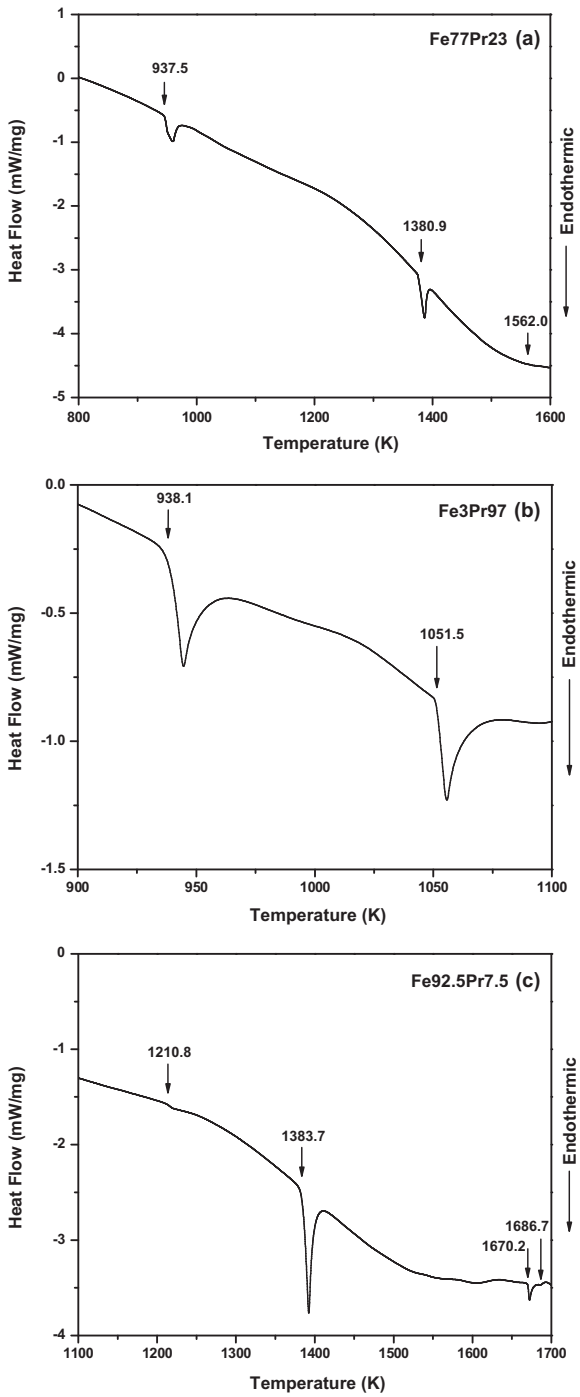

Fig. 1. DTA curves of the annealed Fe-Pr alloys at different heating rate. (a) Fe77Pr23 (10 K/min); (b) Fe3Pr97 (20 K/min); (c) Fe92.5Pr7.5 (20 K/min).

found, which are corresponding to the eutectic reaction (L↔$Fe_{17}Pr_2$+dhcp-Pr) at 937.5 K and the peritectic reaction (L+fcc-Fe↔$Fe_{17}Pr_2$) at 1380.9 K, respectively, and the liquidus temperature was set as 1562.0 K. According to the onset temperatures of the thermal effects in the heating DTA curve of Fe3Pr97 alloy in Fig. 1(b), the eutectic reaction (L↔$Fe_{17}Pr_2$+dhcp-Pr) at 938.1 K and the peritectic reaction (L+bcc-Pr↔dhcp-Pr) at 1051.5 K were determined. In Fig. 1(c), the four peaks in the DTA curve of Fe92.5Pr7.5 alloy are successively related to the peritectic reaction (L+fcc-Fe↔bcc-Fe) at 1210.8 K, the peritectic reaction (L+fcc-Fe↔$Fe_{17}Pr_2$) at 1383.7 K, the peritectic reaction (L+bcc-Fe↔fcc-Fe) at 1670.2 K and the liquidus temperature at 1686.7 K.

Table 1
Experimental results of the Fe-Pr alloys determined by DTA measurements.

Alloy compositions (at%)	Phase transformation temperatures (K)					
	Invariant reactions					Liquidus
Fe92.5Pr7.5			1210.8	1383.7	1670.2	1686.7
Fe77Pr23	937.5			1380.9		1562.0
Fe70Pr30	938.3			1383.0		1508.8
Fe60Pr40	938.1			1383.7		—
Fe53Pr47	938.2					—
Fe42Pr58	938.5					—
Fe33Pr67	938.6					—
Fe25Pr75	938.2					1020.8[a]
Fe12.5Pr87.5	937.5					1015.3[a]
Fe3Pr97	938.1	1051.5				—

[a] Note: The values are referred only because of the weak thermal effect in the DTA curves and the serious oxidation of Fe-Pr alloy with high Pr content at high temperature.

In the present experiments, the temperatures of the eutectic reaction (L↔$Fe_{17}Pr_2$+dhcp-Pr), the peritectic reaction (L+fcc-Fe↔$Fe_{17}Pr_2$) and the peritectic reaction (L+bcc-Pr↔dhcp-Pr) were determined to be 938.5 K, 1383.0 K and 1051.5 K, respectively. In addition, the reaction temperatures, L+bcc-Fe↔fcc-Fe and fcc-Fe+$Fe_{17}Pr_2$↔bcc-Fe, were determined to be 1670.2 K and 1210.8 K, respectively. The phase transformation temperatures of Fe-Pr alloys determined in the present work were shown in Table 1. It should be noted that the liquidus temperatures of Fe60Pr40, Fe53Pr47, Fe42Pr58, Fe33Pr67 and Fe3Pr97 alloys were not determined accurately in this work due to the obvious oxidation of the liquid alloys at high temperature.

3.2. The Fe-Nd binary system

Fig. 2 is the representative DTA curves of Fe-Nd alloys. In Fig. 2(a), there are three peaks in the heating DTA curve (20 K/min) of annealed Fe75Nd25 alloy, which represent to the eutectic reaction (L↔$Fe_{17}Nd_5$+hcp-Nd) at 959.4 K, the peritectic reaction (L+fcc-Fe↔$Fe_{17}Nd_2$) at 1492.4 K and the liquidus temperature at 1563.7 K. It should be noticed that the peritectic reaction (L+$Fe_{17}Nd_2$↔$Fe_{17}Nd_5$) did not occur in the present DTA measurements, which could result from the non-equilibrium microstructure of the as-cast alloy. In Fig. 2(b), the temperature of the eutectic reaction (L↔hcp-Nd +$Fe_{17}Nd_5$) at 958.4 K and liquidus temperature at 978.8 K in the DTA curve of Fe17Nd83 alloy were determined. The four peaks of the heating DTA curve of Fe92.5Nd7.5 alloy in Fig. 2(c) are responsible for the peritectic reactions (fcc-Fe+$Fe_{17}Nd_2$↔bcc-Fe) at 1211.7 K, (L+fcc-Fe↔$Fe_{17}Nd_2$) at 1490.4 K, the peritectic reaction (L+bcc-Fe↔fcc-Fe) at 1670.8 K and liquidus temperture at 1686.0 K. The preitectic reaction (L+$Fe_{17}Nd_2$↔$Fe_{17}Nd_5$) was not observed in the DTA curves, which could result from the insufficient annealed time of the alloy sample to form $Fe_{17}Nd_5$ phase.

In the present work, temperatures of the invariant reactions, L+bcc-Fe↔fcc-Fe at 1670.8 K, L+fcc-Fe↔$Fe_{17}Nd_2$ at 1490.0 K, L↔$Fe_{17}Nd_5$+dhcp-Nd at 958.5 K, L+bcc-Nd↔dhcp-Nd at 1121.2 K and fcc-Fe+$Fe_{17}Nd_2$↔bcc-Fe at 1211.7 K were determined in the present work, respectively. The phase transformation temperatures of Fe-Nd alloys determined in the present work were given in Table 2. Duo to the oxidation of liquid alloys at high temperature, the liquidus temperatures of Fe85Nd15, Fe50Nd50, Fe45Nd55 and Fe33Nd67 alloys were not be determined experimentally in this work.

4. Thermodynamic models

4.1. Pure elements

In the Fe-Pr and Fe-Nd systems, the Gibbs energy of pure element can be given as:

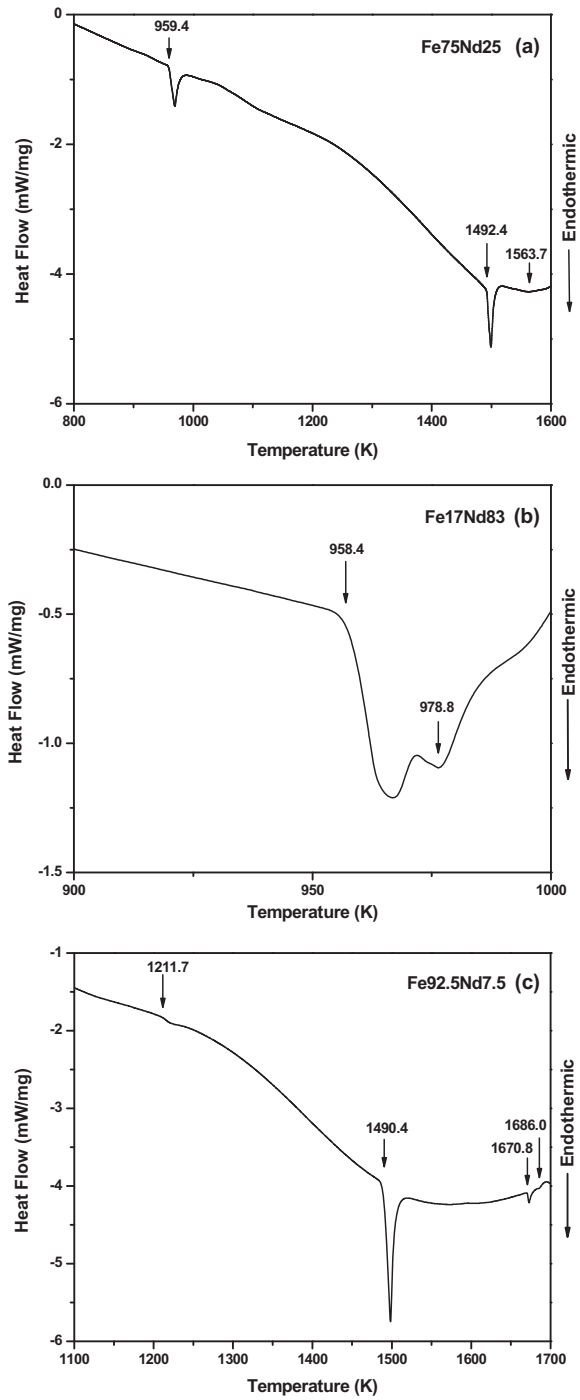

Fig. 2. DTA curves of the annealed Fe-Nd alloys at different heating rate. (a) Fe75Nd25 (20 K/min); (b) Fe17Nd83 (10 K/min); (c) Fe92.5Nd7.5 (20 K/min).

$$^{0}G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

in which $G_i^{\phi}(T)$ stands for the Gibbs energy of the element i with structure ϕ. H_i^{SER} is the enthalpy of the element i in standard reference state (298.15 K, 1 bar). T is the absolute temperature in Kelvin. $^{0}G_i^{\phi}(T)$ is the molar Gibbs energy of element i in structure of ϕ referred to the enthalpy of its stable state at 298.15 K and 1 bar. In this work, the Gibbs energies of the elements, $^{0}G_{Fe}^{\phi}(T)$, $^{0}G_{Pr}^{\phi}(T)$ and $^{0}G_{Nd}^{\phi}(T)$, are taken from the Scientific Group Thermodata Europe (SGTE) compiled by Dinsdale [51].

Table 2
Experimental results of the Fe-Nd alloys determined by DTA measurements.

Alloy compositions (at%)	Phase transformation temperature (K)					
	Invariant reactions					liquidus
Fe92.5Nd7.5			1211.7	1490.4	1670.8	1686.0
Fe85Nd15	962.9			1489.3		—
Fe75Nd25	959.4	—		1492.4		1563.7
Fe65Nd35	960.2	—				1480.7
Fe60Nd40	958.0	—				1470.7
Fe50Nd50	960.2	—				—
Fe45Nd55	958.0	—				—
Fe33Nd67	957.4					—
Fe17Nd83	958.4					978.8[a]
Fe3Nd97	959.3	1121.2				1210.6[a]

[a] Note: The values are referred only because of the weak thermal effect in the DTA curves and the serious oxidation of Fe-Nd alloy with high Nd content at high temperature.

4.2. Solution phases

Based on random mixing of the constituent atoms, the solution phase ϕ (ϕ=liquid, bcc-Fe, fcc-Fe, dhcp-Nd, dhcp-Pr) is described with the substitutional solution model, and the Gibbs energies of the solution phases can be expressed as follows:

$$G_m^{\phi} = \sum_{i=Fe,RE} {}^{0}G_i^{\phi} x_i + RT \sum_{i=Fe,RE} x_i \ln x_i + {}^{ex}G_m^{\phi} + {}^{mag}G_m^{\phi} \quad (2)$$

$$^{ex}G_m^{\phi} = x_{Fe}x_{RE} \sum_{j=0}^{n} {}^{j}L_{Fe,RE}^{\phi}(x_{Fe} - x_{RE})^j \quad (3)$$

$$^{mag}G_m^{\phi} = RT\ \ln(\beta_0 + 1)g(\tau) \quad (4)$$

which x_i means the mole fraction of the element i, T stands for the absolute temperature in Kelvin, and R is the gas constant. τ is the normalized temperature as expressed, $\tau = T/T_c^{\phi}$, and T_c^{ϕ} is the Curie temperature of phase ϕ. β_O is the Bohr magnetrons. $^{ex}G_m^{\phi}$ is the excess Gibbs energy. $^{mag}G_m^{\phi}$ is the magnetic contribution to the Gibbs energy of phase ϕ. $^{j}L_{Fe,RE}^{\varphi}$ is the binary interaction parameters and can be expressed as:

$$^{j}L_{Fe,RE}^{\phi} = A_j + B_jT \quad (5)$$

where A_j and B_j are the parameters to be optimized in this work. In addition, according the equation suggested by Hillert and Jarl [52], $g(\tau)$ is expressed as:

$$g(\tau) = 1 - \frac{1}{M}\left[\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p} - 1\right)\left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600}\right)\right] \quad (\leq 1) \quad (6)$$

$$g(\tau) = -\frac{1}{M}\left(\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500}\right) \quad (>1) \quad (7)$$

$$M = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p} - 1\right) \quad (8)$$

in which the parameter p is determined by the structure of the phase ϕ (0.4 for bcc phase and 0.28 for other phases).

4.3. Intermetallic compounds

The intermetallic compounds, $Fe_{17}Pr_2$, $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$ are treated as the stoichiometric compounds in this work. Because the heat capacity of $Fe_{17}Pr_2$ was experimentally determined by Mandal et al. [35], the Gibbs energy of $Fe_{17}Pr_2$ can be expressed as:

Table 3
Thermodynamic parameters of the Fe-Pr binary system.

Phases	Thermodynamic parameters[a]	Reference
Liquid	${}^{0}L^{liq}_{Fe,Pr} = 635.245 + 6.87T$	This work
	${}^{1}L^{liq}_{Fe,Pr} = -5681.778 + 9.6T$	
	${}^{2}L^{liq}_{Fe,Pr} = 1373.68 - 2T$	
$Fe_{17}Pr_2$	$G_m^{Fe_{17}Pr_2} = -174947.1 + 411.43T - 35T\ln T - 0.0355T^2 + 17{}^{0}G^{bcc}_{Fe} + 2{}^{0}G^{dhcp}_{Pr}$	This work
	$T_C^{Fe_{17}Pr_2} = 283$	[13]
	$\beta_0 = 30.6$	[34]
bcc	${}^{0}L^{bcc}_{Fe,Pr} = 100000$	This work
fcc	${}^{0}G^{fcc}_{Pr} = 50000 + {}^{0}G^{dhcp}_{Pr}$	This work
	${}^{0}L^{fcc}_{Fe,Pr} = 40000$	
dhcp	${}^{0}G^{dhcp}_{Pr}$ cited from SGTE database	[51]

[a] Note: The Gibbs energy (J/mol) of Fe and Pr with liquid, fcc, bcc and dhcp structures are given by Dinsdale [51]. The unit of temperature (T) is Kelvin (K).

Table 4
Thermodynamic parameters of the Fe-Nd binary system.

Phases	Thermodynamic parameters[a]	Reference
Liquid	${}^{0}L^{liq}_{Fe,Nd} = -4103.5 + 7.7T$	This work
	${}^{1}L^{liq}_{Fe,Nd} = -1883.84 + 6.8T$	
	${}^{2}L^{liq}_{Fe,Nd} = 153.74 + 1.65T$	
$Fe_{17}Nd_2$	$G_m^{Fe_{17}Nd_2} = -165000 + 89.542T + 17{}^{0}G^{bcc}_{Fe} + 2{}^{0}G^{dhcp}_{Nd}$	This work
	$T_C^{Fe_{17}Nd_2} = 327$	[14]
	$\beta_0 = 2.3$	[14]
$Fe_{17}Nd_5$	$G_m^{Fe_{17}Nd_2} = -165000 + 83.888T + 17{}^{0}G^{bcc}_{Fe} + 5{}^{0}G^{dhcp}_{Nd}$	This work
	$T_C^{Fe_{17}Nd_5} = 503$	[44]
	$\beta_0 = 2.3$	[43]
bcc	${}^{0}L^{bcc}_{Fe,Nd} = 72000$	This work
fcc	$G^{fcc}_{Nd} = 20000 + {}^{0}G^{dhcp}_{Nd}$	This work
	${}^{0}L^{fcc}_{Fe,Nd} = 50000$	
dhcp	${}^{0}G^{dhcp}_{Nd}$ cited from SGTE database	[51]

[a] Note: The Gibbs energy (J/mol) of Fe and Nd with liquid, fcc, bcc and dhcp structures are given by Dinsdale [51]. The unit of temperature (T) is Kelvin (K).

$$G_m^{Fe_{17}Pr_2} = 17\ {}^{0}G^{bcc}_{Fe} + 2\ {}^{0}G^{dhcp}_{Pr} + C + DT + ET\ln(T) + FT^2 + {}^{mag}G_m^{Fe_{17}Pr_2} \quad (9)$$

Due to the lack of experimental heat capacity data, using the Neumann-Kopp rule, the Gibbs energies of $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$ are described as:

$$G_m^{Fe_{17}Nd_2} = 17\ {}^{0}G^{bcc}_{Fe} + 2\ {}^{0}G^{dhcp}_{Nd} + I + JT + {}^{mag}G_m^{Fe_{17}Nd_2} \quad (10)$$

$$G_m^{Fe_{17}Nd_5} = 17\ {}^{0}G^{bcc}_{Fe} + 5\ {}^{0}G^{dhcp}_{Nd} + M + NT + {}^{mag}G_m^{Fe_{17}Nd_5} \quad (11)$$

The parameters C, D, E, F, I, J, M and N, are to be optimized in this work. The magnetic contribution (${}^{mag}G_m^{Fe_{17}Pr_2}$, ${}^{mag}G_m^{Fe_{17}Nd_2}$ and ${}^{mag}G_m^{Fe_{17}Nd_5}$) were expressed by Eq. (4). In this work, the Curie temperature (T_c) and Bohr magnetrons (β_O) of three intermetallic compounds, $Fe_{17}Pr_2$, $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$, are 283 K and 30.6 μ_B, 327 K and 2.3 μ_B, 503 K and 2.3 μ_B [13,14,41,44], respectively.

5. Results and discussion

Based on the experimental data determined in this work and reported in the literature as well as the stabilities of the elements Fe, Pr and Nd compiled by Dinsdale [52], the model parameters of various phases in the Fe-Pr and Fe-Nd binary systems were optimized using the PARROT module in the Thermo-calc® software package developed by Sundman et al. [53]. The PARROT module works by minimizing the square sum of the differences between the experimentally determined data and the calculated values. During the optimizing procedure, each set of the experimental data is given a certain weight according to the reliability and compatibility of the experimental data. It should be mentioned that the more reliable the experimental data are higher weight to be set. The weight of each experimental data can be changed

Table 5
Invariant reactions in the Fe-Pr binary system.

Invariant reactions	Type	Temperature (K)	Composition (x^L_{Pr})	Reference
Liq.+δ-Fe↔γ-Fe	peritectic	1671.5	0.0895	This work (cal.)
		1670.2	—	This work (exp.)
		1588	—	[20]
		1671 ± 5	—	[13]
Liq.+γ-Fe↔ $Fe_{17}Pr_2$	peritectic	1383.0	0.470	This work (cal.)
		1383.0	—	This work (exp.)
		1388.0	0.479	[20]
		1378 ± 8	—	[13]
		1371	—	[16]
		1375	—	[22]
		1378	0.45	[25]
		1381	—	[33]
Liq.↔ $Fe_{17}Pr_2$+α-Pr	eutectic	939.3	0.802	This work (cal.)
		938.5	—	This work (exp.)
		922.3	0.80	[20]
		940 ± 3	—	[13]
		939	—	[16]
		940	0.82	[22]
Liq.+β-Pr↔α-Pr	peritectic	1068.0	0.924	This work (cal.)
		1051.5	—	This work (exp.)
		1064	—	[20]
		1054 ± 6	—	[13]
		1068	—	[22]
γ-Fe+$Fe_{17}Pr_2$ ↔α-Fe	peritectic	1184.6	—	This work (cal.)
		1210.8	—	This work (exp.)
		1210	—	[20]
		1189 ± 10	—	[13]
		1183	—	[22]

Table 6
Invariant reactions in the Fe-Nd binary system.

Invariant reactions	Type	Temperature (K)	Composition (x^L_{Nd})	Reference
Liq.+δ-Fe↔γ-Fe	peritectic	1670.9	0.105	This work (cal.)
		1670.8	—	This work (exp.)
		1646	—	[20]
		1665	—	[37]
		1667	—	[41]
		1643	—	[38]
Liq.+γ-Fe↔ $Fe_{17}Nd_2$	peritectic	1490.0	0.340	This work (cal.)
		1490.0	—	This work (exp.)
		1478	0.315	[14]
		1482.4	—	[17]
		1488.9	0.272	[20]
		1483	—	[33]
		1463	—	[37]
		1483	—	[38]
		1483	—	[39]
Liq.+$Fe_{17}Nd_2$↔ $Fe_{17}Nd_5$	peritectic	1071.0	0.711	This work (cal.)
		1044	0.70	[14]
		1071.5	—	[17]
		1071.3	0.695	[20]
		1068	—	[41]
		1071	—	[47]
Liq.↔ $Fe_{17}Nd_5$+β-Nd	eutectic	958.4	0.783	This work (cal.)
		958.5	—	This work (exp.)
		956	—	[14]
		955.45	0.783	[17]
		947.6	0.775	[20]
		920	—	[37]
		958	—	[38]
		957.7 ± 0.5	0.765	[39]
Liq.+α-Nd↔β-Nd	peritectic	1127.7	0.916	This work (cal.)
		1121.2	—	This work (exp.)
		1127	0.909	[14]
		1121	—	[17]
		1121	—	[20]
		1133	—	[37]
		1122 ± 1	—	[39]
γ-Fe+$Fe_{17}Nd_2$↔ α-Fe	peritectic	1184.6	—	This work (cal.)
		1211.7	—	This work (exp.)
		1208	—	[14]
		1183.5	—	[17]
		1184		[20]
		1208	—	[37]

systematically during the optimization until most of the experimental data can be taken into consideration within the claimed uncertainty limits.

Thermodynamic parameters of all phases obtained finally in the Fe-Pr and Fe-Nd binary systems were summarized in Tables 3 and 4, respectively. Using the present thermodynamic parameters, the temperatures and compositions of the invariant reactions in the Fe-Pr and Fe-Nd binary systems were calculated as shown in Table 5 and 6, respectively. The comparisons of the calculated results with the experimental data were discussed in detail in the following section.

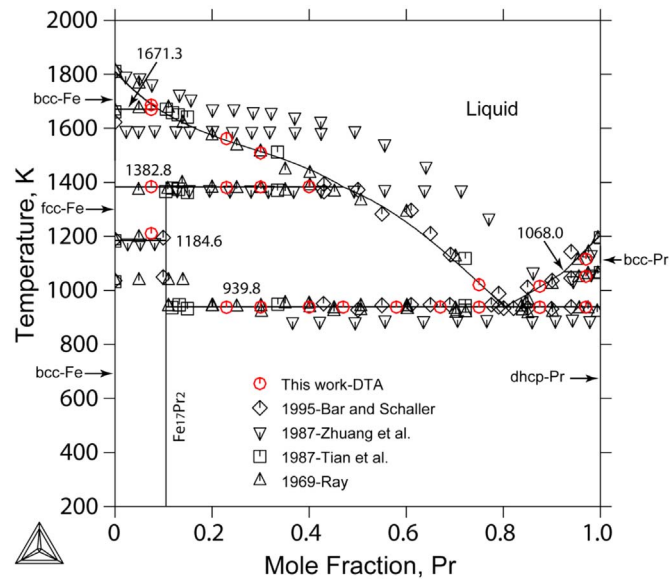

Fig. 3. Calculated Fe-Pr phase diagram with the experimental data determined in this work and in Refs. [13,21–23]..

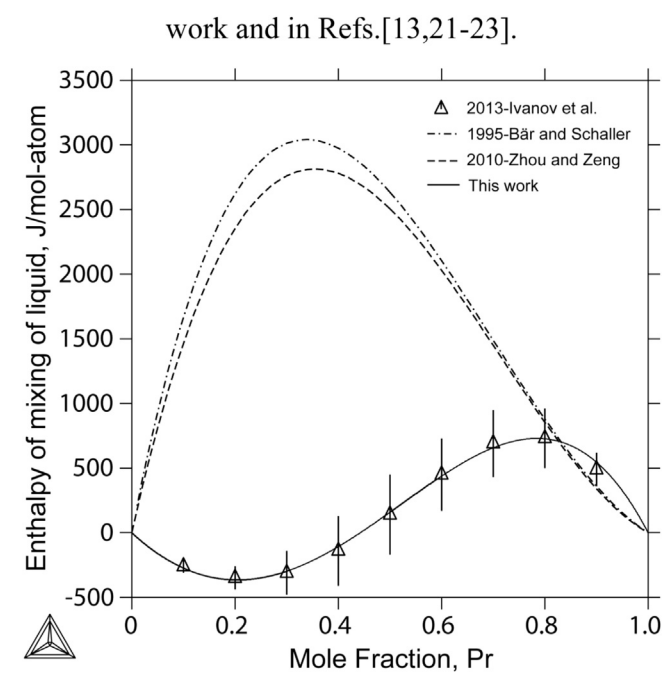

Fig. 4. Calculated enthalpy of mixing of Fe-Pr liquid alloys at 1829 K with the experimental data [19] and calculated results [13,16]. (Reference states: liquid Fe and Pr).

5.1. The Fe-Pr binary system

Fig. 3 is the calculated Fe-Pr binary phase diagram with the experimental data determined in this work and reported by Bär and Schaller [13], Ray [21], Tian et al. [22] and Zhuang et al. [23]. The calculated liquidus of the Fe-Pr binary system shows the good agreement with the experimental data determined in the present work and in Refs. [13,21,22], while it is the obvious difference between the data reported by Zhuang et al. [23]. The reason for the difference could be resulted from the mistakes of the unit of experimental data as given in the figure by Zhuang et al. [23], which was pointed out by Konar et al. [20]. As can be seen in Table 5, the calculated temperatures of the eutectic reaction (L↔$Fe_{17}Pr_2$+dhcp-Pr) and the peritectic reactions (L +fcc-Fe↔$Fe_{17}Pr_2$, L+fcc-Fe↔bcc-Fe) are 939.3 K, 1383.0 K and

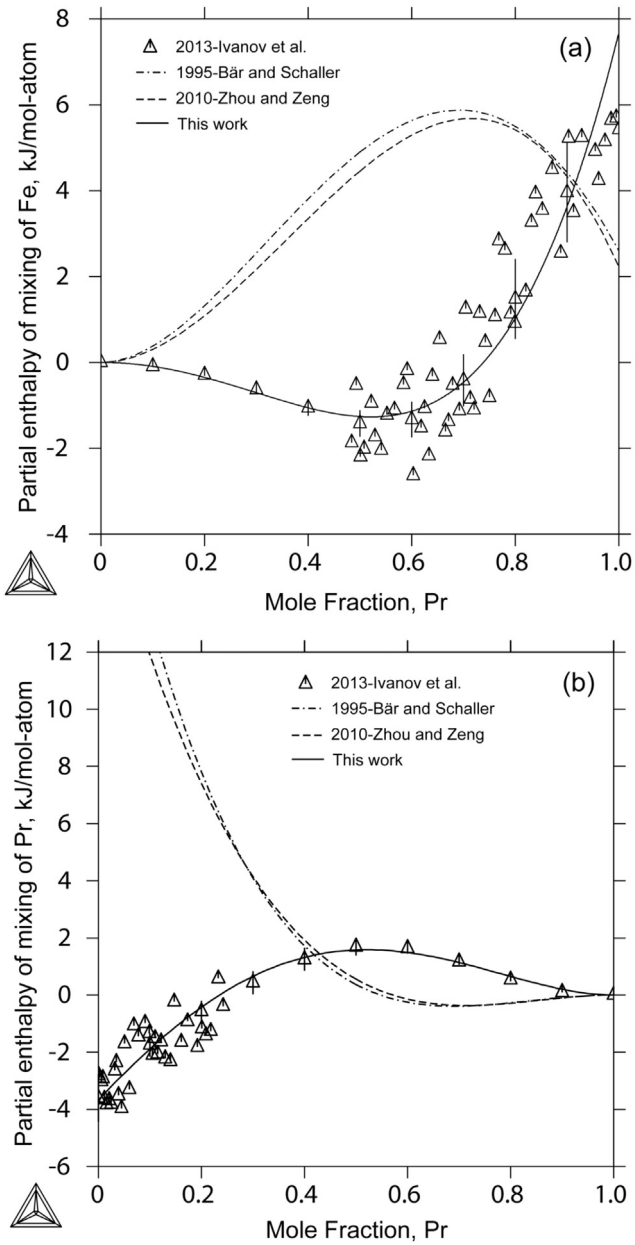

Fig. 5. Calculated partial molar enthalpies of mixing of Fe (a) and Pr (b) in the Fe-Pr liquid alloys at 1829 K with the experimental data [19] and the calculated results [13,16]. (Reference states: liquid Fe and Pr).

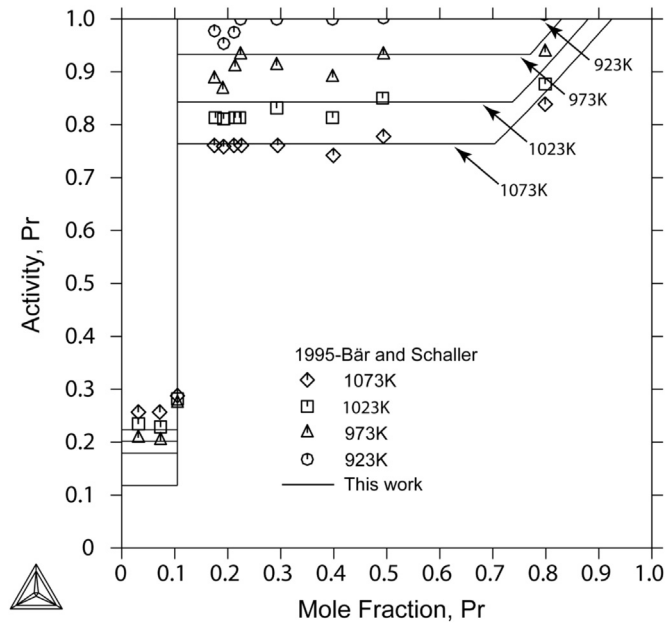

Fig. 6. Calculated activities of Pr at different temperatures with the experimental data [13]. (Reference state: dhcp-Pr (923 K, 973 K, 1023 K), bcc-Pr (1073 K)).

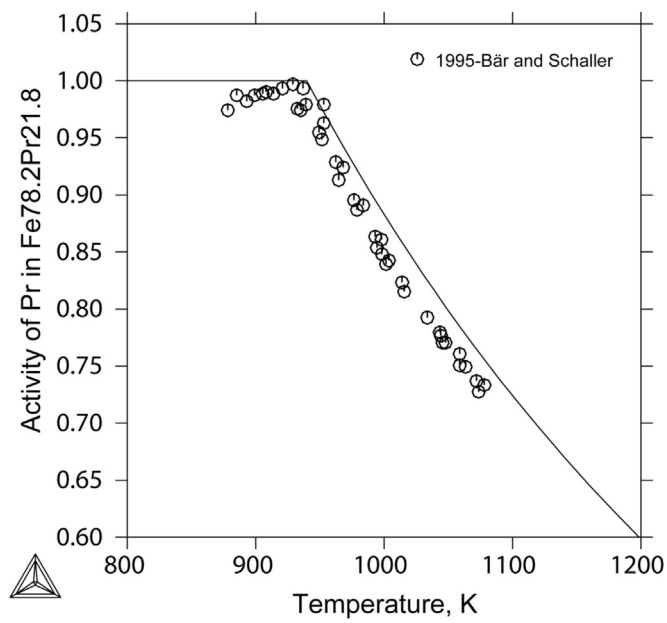

Fig. 7. Calculated temperature dependence of the activities of Pr in Fe78.2Pr21.8 alloy with the experimental data [13]. (Reference state: dhcp-Pr).

1671.5 K, respectively, which are in good agreement with the experimental data determined in this work (938.5 K, 1383.0 K and 1670.2 K) and the reported data in Refs. [13,22]. The calculated temperature of the invariant reaction (bcc-Fe+$Fe_{17}Pr_2$↔fcc-Fe) at 1184.6 K is lower than the experimental data (1189 ± 10 K) [13] and the data (1210.8 K) determined in this work. Besides, the calculated temperature of the invariant reaction, L+bcc-Pr↔dhcp-Pr, at 1068.0 K, is different with the experimental value (1051.5 K) determined in this work and the data (1054 ± 6 K) reported by Bär and Schaller [13]. The reason for it is that the solubility of Fe in dhcp-Pr (α-Pr) and bcc-Pr (β-Pr) as well as the solubility of Pr in bcc-Fe (α-Fe, δ-Fe) and fcc-Fe (γ-Fe) were not taken into consideration in the present work.

Fig. 4 is the calculated enthalpy of mixing of liquid Fe-Pr alloys at 1829 K with the experimental data measured by Ivanov et al. [19] and calculated results [13,16]. Within experimental error, the calculated results in the present work are in good agreement with the experimental data [19], while it is obvious difference between the calculated results in Refs [13,16] and the experimental data [19]. Fig. 5 is the partial enthalpies of mixing of Fe and Pr in liquid Fe-Pr alloys at 1829 K. The calculated results show good agreement with the experimental data [19].

Fig. 6 is the comparison of the calculated activities of Pr in the Fe-Pr alloys at 923 K, 973 K, 1023 K and 1073 K with the experimental results [13]. The calculated results can reproduce well the experimental results at the different temperatures. Fig. 7 shows the calculated temperature dependence of the activities of Pr in Fe78.2Pr21.8 alloy with the experimental data reported by Bär and Schaller [13]. The calculated activities of Pr are slightly higher than the experimental data in high temperature range (> 940 K).

Fig. 8 is the calculated heat capacity of $Fe_{17}Pr_2$ with the experimental data reported by Mandal et al. [35] and the calculated results by Bär and Schaller [13] and Zhou and Zeng [16]. As can be seen, the

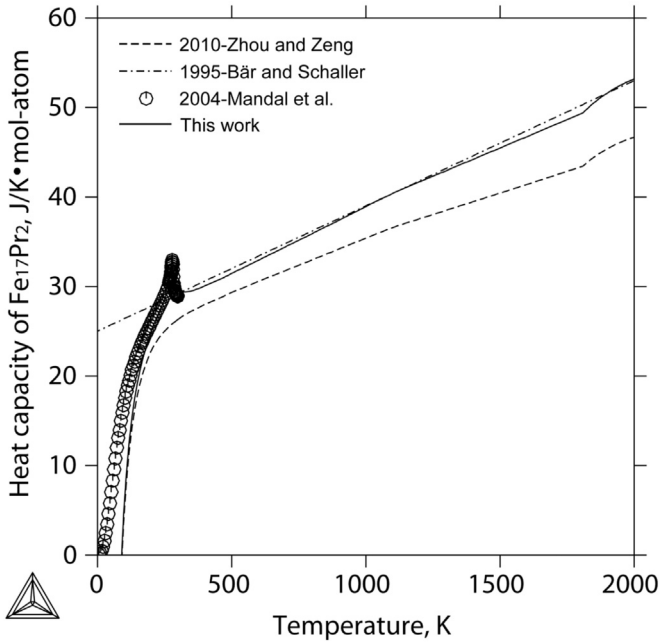

Fig. 8. Calculated heat capacity of $Fe_{17}Pr_2$ with the experimental data [35] and calculated results [13,16].

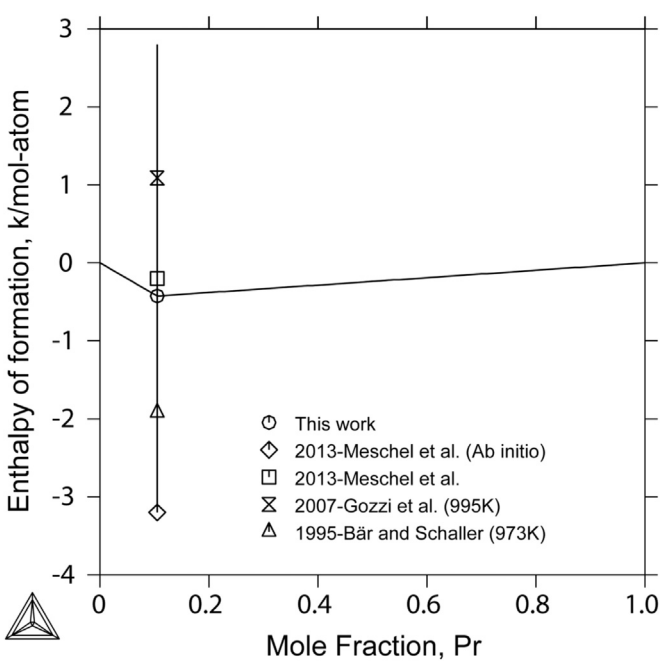

Fig. 9. Calculated enthalpy of formation of $Fe_{17}Pr_2$ with the experimental data [13,33,34]. (Reference states: bcc-Fe and dhcp-Pr).

calculated results in the present work are consistent with the experimental data [35] and match well with the calculated results [13] above the Curie temperature. In addition, as shown in Fig. 9 and Table 7, the enthalpy of formation of $Fe_{17}Pr_2$ was calculated to be −426 J/mol-atom at 298 K. The calculated value agrees with the experimental data [13,33], but is obviously different with the positive value determined by Gozzi et al. [34]. Normally, the enthalpy of formation of the stable intermetallic compound at room temperature is negative. Therefore, the present calculated result is reasonable.

5.2. The Fe-Nd binary system

Fig. 10 shows the calculated phase diagram of Fe-Nd binary system

Table 7
Enthalpy of formation of intermetallic compounds in Fe-Pr and Fe-Nd binary systems.

Intermetallic compounds	Enthalpy of formation (298 K) (J/mol-atom)	Method	Reference
$Fe_{17}Pr_2$	−426	Calculation	This work
	−1920 (973 K)	EMF	[13]
	1973	Calculation	[20]
	−200 ± 3000	EMF	[33]
	−3200	Ab initio	[33]
	1090 (995 K)	EMF	[34]
$Fe_{17}Nd_2$	−593	Calculation	This work
	530	Calculation	[20]
	−3000 ± 3900	EMF	[33]
	−3800	Ab initio	[33]
	−640 ± 10	EMF	[34]
$Fe_{17}Nd_5$	−610	Calculation	This work
	523	Calculation	[20]

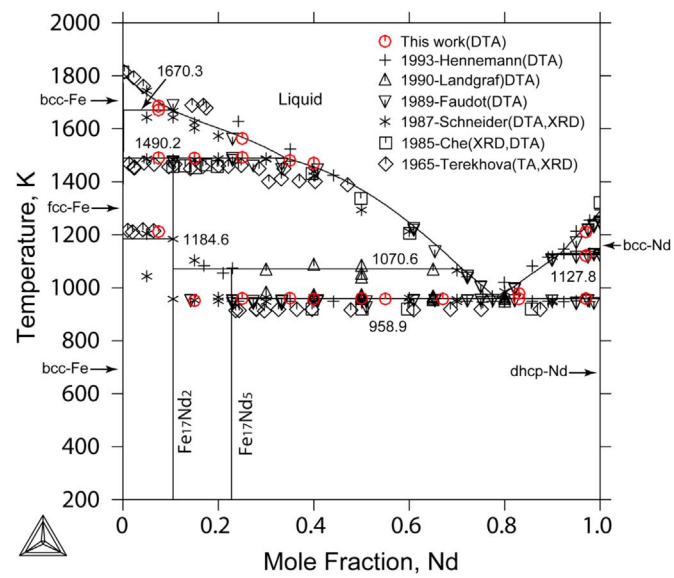

Fig. 10. Calculated Fe-Nd phase diagram with the experimental data determined in this work and in Refs. [14,36–39,41].

with the experimental data determined in this work and in Refs. [14,36–39,41]. As can be seen, the calculated results are in good agreement with the experimental data measured in the present work and in Refs. [14,38,39,41], while show obviously difference with the experimental data reported by Terekhova et al. [36] and Che et al. [37]. It should be noted that $Fe_{17}Nd_2$ as the only stable intermetallic compound was reported by Che et al. [33]. In addition, the experimental data [36,37] are obviously lower than those determined in Refs. [14,38,39,41], which were given lower weight in the optimization process in this work.

The calculated temperatures and compositions of the invariant reactions in the Fe-Nd binary system with the experimental data were summarized in Table 6. The calculated temperatures of the eutectic reaction (L↔$Fe_{17}Nd_5$+hcp-Nd at 958.4 K) and the peritectic reaction (L+fcc-Fe↔$Fe_{17}Nd_2$ at 1490.0 K) are in good agreement with the experimental data determined in this work and the reported data [39] within experimental error, while the calculated temperature of the peritectic reaction (L+$Fe_{17}Nd_2$↔$Fe_{17}Nd_5$ at 1070.0 K) is a little higher (3 K) than the experimental data [33]. The temperature of the peritectic reaction (L+bcc-Fe↔fcc-Fe at 1670.9 K) is in good agreement with the experimental data measured in this work and a little higher (5.9 K) than the experimental data [37]. The calculated temperature of the peritectic reactions (L+bcc-Nd↔hcp-Nd at 1127.7 K

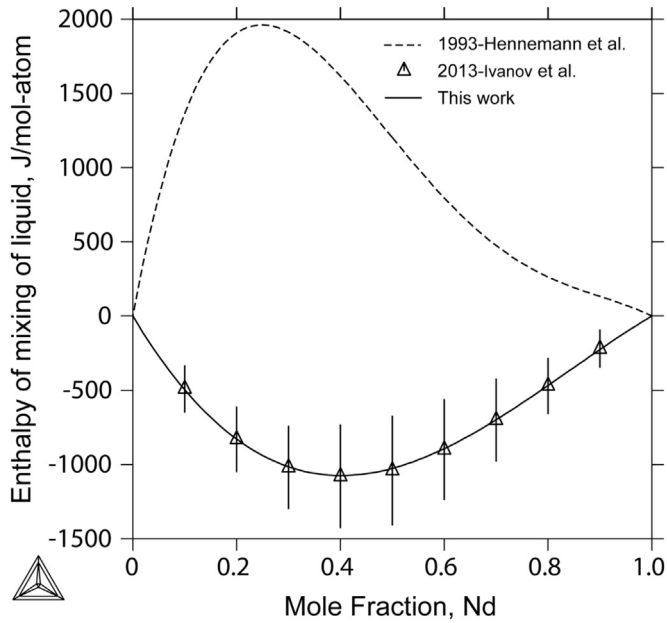

Fig. 11. The calculated enthalpy of mixing of Fe-Nd liquid alloys at 1829 K with the experimental data [19] and the calculated results [14]. (Reference states: liquid Fe and liquid Nd).

and fcc-Fe+$Fe_{17}Nd_2$↔bcc-Fe at 1184.6 K) are obviously different with the experimental data determined in this work and the reported data [37,39]. The reason for this difference could be that the solubility of Fe in bcc-Nd (α-Nd) and dhcp-Nd (β-Nd) and the solubility of Nd in bcc-Fe (α-Fe, δ-Fe) and fcc-Fe (γ-Fe) were not considered in the present work.

Fig. 11 is the comparison of the calculated enthalpy of mixing of liquid Fe-Nd alloys at 1829 K with the experimental data determined by Ivanov et al. [19] and calculated results by Hennmann et al. [14]. The calculated present results in the whole composition range are in good agreement with the experimental data [19], but shows obvious difference with the calculation [14]. Fig. 12 is the partial molar enthalpy of mixing of Fe and Nd in liquid alloys at 1829 K. The present calculation can match well the experimental data measured by Ivanov et al. [19], but is obvious difference with the previous calculation [14].

Fig. 13 is the calculated activities of Nd in the Nd-Fe alloys at 973 K, 1073 K, 1373 K, 1398 K, 1423 K, 1448 K, 1473 K, 1498 K and 1523 K with the experimental data reported by Hennmann et al. [14] and Nagai et al. [50]. As can be seen in Fig. 13(a), the calculated activities curves agree well with the experimental results [14] except for the slightly lower in the Fe-$Fe_{17}Nd_2$ region. In Fig. 13(b), the calculation show obviously higher than the experimental data determined by means of the double Knudsen cell mass spectrometry [50]. During the optimization, the attempt to fit these activities data [50] would make it impossible to reproduce the Fe-Nd binary phase diagram. Therefore, the calculated activities are reasonable in the present work.

The comparison of the calculated enthalpy of formation of $Fe_{17}Nd_2$ (−593 J/mol-atom) and $Fe_{17}Nd_5$ (−610 J/mol-atom) at 298 K with the experimental data [33,34] is shown in Fig. 14 and Table 7. The Gibbs energies of formation of $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$ at 973 K were calculated to be −48.32 kJ/mol-atom and −53.37 kJ/mol-atom, respectively. The calculated results agree well with the experimental results (−46 kJ/mol-atom and −52 kJ/mol-atom at 973 K) reported by Hennmann et al. [14], and higher than the experimental value (−36.9 ± 0.02 kJ/mol-atom for $Fe_{17}Nd_2$ at 1022 K) reported by Gozzi et al. [34].

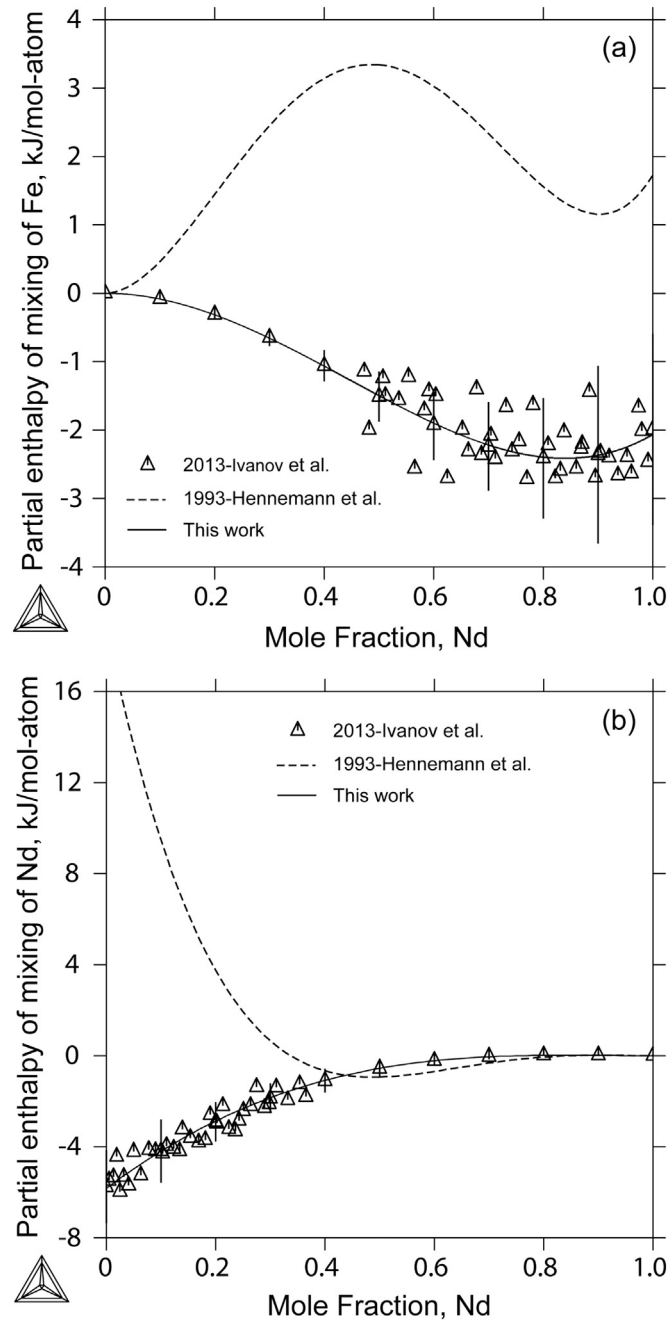

Fig. 12. Calculated partial molar enthalpies of mixing of Fe (a) and Nd (b) in liquid Fe-Nd alloys at 1829 K with the experimental data [19] and the calculated results [14]. (Reference states: liquid Fe and Nd).

6. Conclusions

The phase diagrams of the Fe-Pr and Fe-Nd binary system were measured with ten alloys prepared by arc-melting method using differential thermal analysis. The temperatures of the invariant reactions and liquidus in the two binary systems were determined in this work, which are in good agreement with the available experimental data. The Fe-Pr and Fe-Nd binary systems were thermodynamically optimized on the bases of experimental thermodynamic data and phase diagram data obtained in this work and in the published literature. The self-consistent thermodynamic parameters have been obtained finally. The calculated Fe-Pr and Fe-Nd phase diagrams and thermodynamic properties agree well with the experimental data.

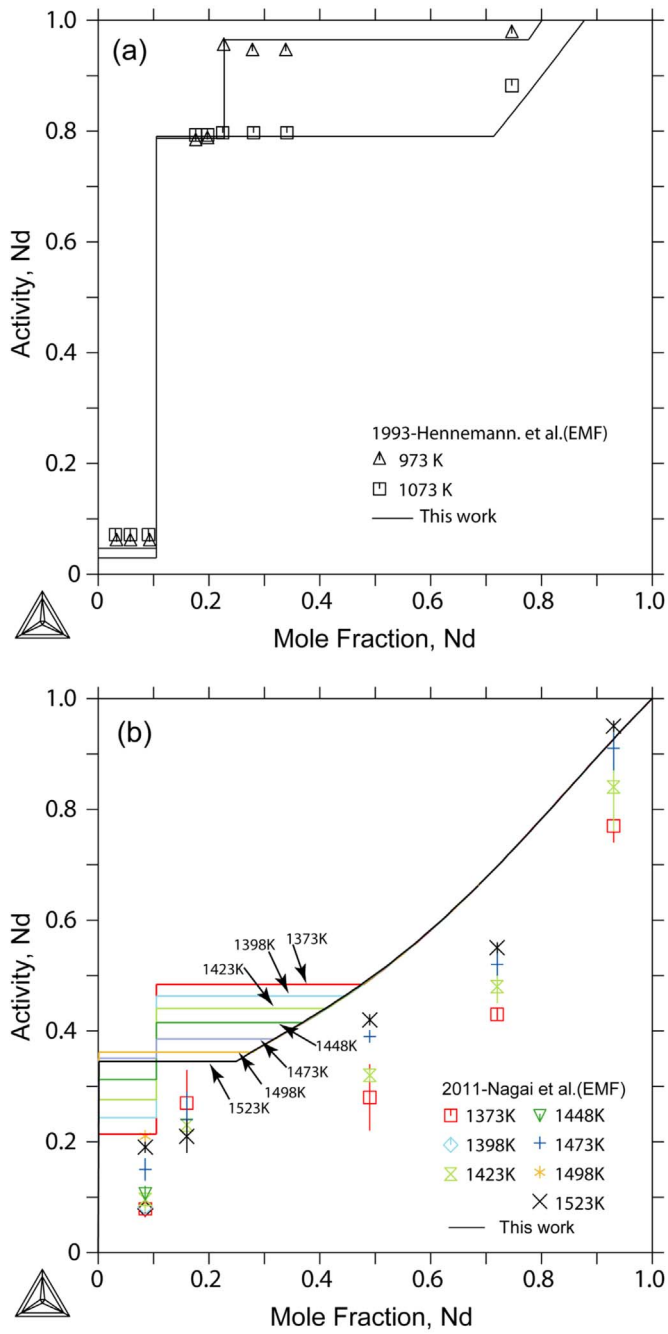

Fig. 13. Calculated activities of Nd in Nd-Fe alloys at different temperatures with the experimental data [14,50]. (a) 973 K and 1073 K (Reference state: dhcp-Nd); (b) 1373 K, 1398 K, 1423 K, 1448 K, 1473 K, 1498 K and 1523 K (Reference state: liquid Nd).

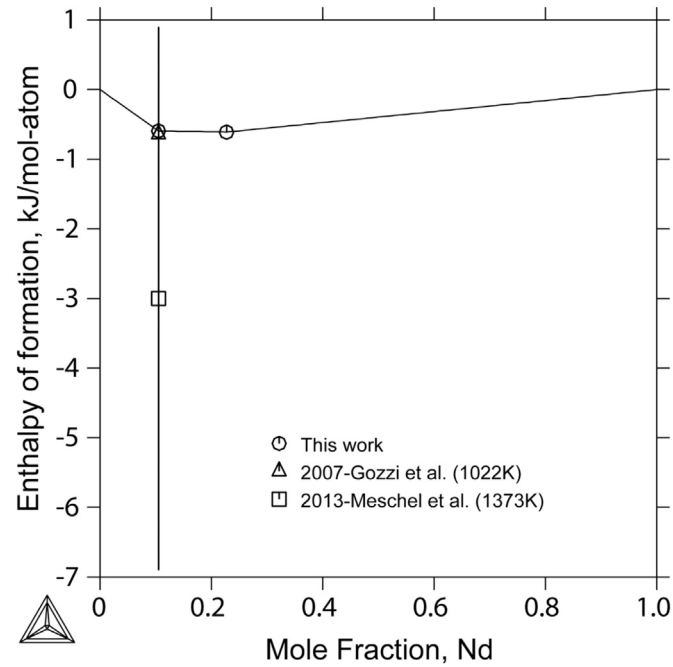

Fig. 14. Calculated enthalpy of formation of $Fe_{17}Nd_2$ and $Fe_{17}Nd_5$ at 298 K with the experimental data [33,34]. (Reference states: bcc-Fe and dhcp-Nd).

Acknowledgements

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2016.10.004.

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